

mTSP Project: Design Pattern Analysis and System Architecture

In this project, the Strategy Design Pattern was applied in the Heuristic Optimization phase to control the five different move operations needed for the mTSP solver. This pattern allows us to define a family of algorithms, encapsulate each one and make them interchangeable. In order to have a consistent contract among all concrete strategies, we created a common interface (MoveStrategy) that declares the apply and getName methods. The five move types were developed as concrete strategy classes implementing this interface : SwapNodesInRoute, SwapHubWithNodeInRoute, SwapNodesBetweenRoutes, InsertNodeInRoute and InsertNodeBetweenRoutes. The MoveContext class is the main context which holds a list of these strategies and decides which one to use during the optimization iterations. With this pattern the project can adhere to the Open/Closed Principle, allowing the addition of new move operations without altering the core optimization logic. Moreover, it enhances maintainability and code readability by encapsulating complex operation logic into distinct classes, thereby preventing convoluted conditional structures and advocating a modular software architecture appropriate for sophisticated academic projects.

